UNIT 2 · TOPIC 2.11 · THE NORMAL DISTRIBUTION

The Normal Distribution

Continuous random variables, the N(μ, σ) family, standardisation and z-scores, the 68–95–99.7 rule, areas under the curve, and inverse-normal calculations for percentiles and interval boundaries.

1. The Normal Curve and Its Parameters

A **continuous** random variable can take any value in an interval, and every interval has a probability. Many such variables are well modelled by a normal distribution: a continuous, unimodal, bell-shaped and symmetric curve identified by two parameters.

NOTATION AND SHAPE

$X \sim N(\mu, \sigma)$ total area under the curve $= 1$

μ locates the centre (and the median and mode); σ sets the spread. A **smaller** σ gives a taller, more concentrated curve; a **larger** σ gives a shorter, flatter one. For a continuous variable $P(X = x) = 0$, so $P(X < x) = P(X \le x)$ — only intervals have positive probability.

2. Z-Scores, Standardisation and the Empirical Rule

STANDARDISATION

$z = (x - \mu) / \sigma$ $Z \sim N(0, 1)$

A z-score is the number of standard deviations a value lies above (positive) or below (negative) the mean. The **standard normal distribution** has $\mu = 0$ and $\sigma = 1$; standardising lets one table handle every normal curve.

EMPIRICAL RULE (68–95–99.7)	INTERVAL	APPROXIMATE AREA
Within 1 SD	$\mu \pm 1\sigma$ (z from −1 to 1)	0.68
Within 2 SD	$\mu \pm 2\sigma$ (z from −2 to 2)	0.95
Within 3 SD	$\mu \pm 3\sigma$ (z from −3 to 3)	0.997

WORKED EXAMPLE

If $X \sim N(70, 5)$, then $x = 80$ has $z = (80 - 70)/5 = 2$: 80 is 2 standard deviations above the mean. By the empirical rule about 95% of values lie between 60 and 80, so roughly 2.5% lie above 80.

3. Areas, Intervals and Inverse Normal Calculations

TWO DIRECTIONS OF CALCULATION

value(s) → area: $P(a < X < b) = \text{normCdf}(a, b, \mu, \sigma)$

area → value: $x = \text{invNorm}(\text{area to the left}, \mu, \sigma)$

Use a very large upper bound such as `9E999` for an unbounded right tail, and a very large negative lower bound (`-9E999`) for a left tail. The lowest $p\%$ corresponds to $\text{invNorm}(p/100, \mu, \sigma)$; the highest $p\%$ uses area $1 - p/100$. `normPdf(x,μ,σ)` returns the **density** (curve height), not a probability — never use it as an answer for "probability".

CALCULATOR COMMANDS (REAL TI-NSPIRE NAMES)

- `normCdf(65,80,70,5)` — $P(65 < X < 80)$ for $N(70,5)$.
- `normCdf(80,9E999,70,5)` — upper tail $P(X > 80)$.
- `invNorm(0.90,70,5)` — the 90th percentile (lowest 90% cut-off).
- `invNorm(0.025,0,1)` — $z^* \approx -1.96$; the middle 95% lies between ± 1.96.
- `randNorm(70,5,100)` — 100 simulated draws from $N(70,5)$.

4. Exam Traps

TRAPS THE EXAM ACTUALLY TESTS

- **State the distribution first**: $X \sim N(\mu, \sigma)$ with units and context, and sketch / shade the region before computing.
- **Probability is area, not height**: `normPdf` is not a probability.
- **"Lowest 10%" vs "highest 10%"**: the latter needs $\text{invNorm}(0.90, \mu, \sigma)$, not 0.10.
- **Middle p%** needs two tails: $\alpha/2 = (1 - p)/2$ in each tail.
- **Empirical rule is approximate** ("about 95%"), while `normCdf` / `invNorm` give exact values — say "approximately" when using 2 SD.
- **Always standardise with the right σ** — a sampling distribution uses the standard error $\sigma/\sqrt{n}$, not the population SD.